

# Naming Unit Test Methods

Although any name is technically allowed, descriptive naming greatly improves clarity

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# Common Naming Pattern

- A widely used convention:

**test<method>\_<condition>\_<expectedResult>**

- Structure improves readability and self-documentation

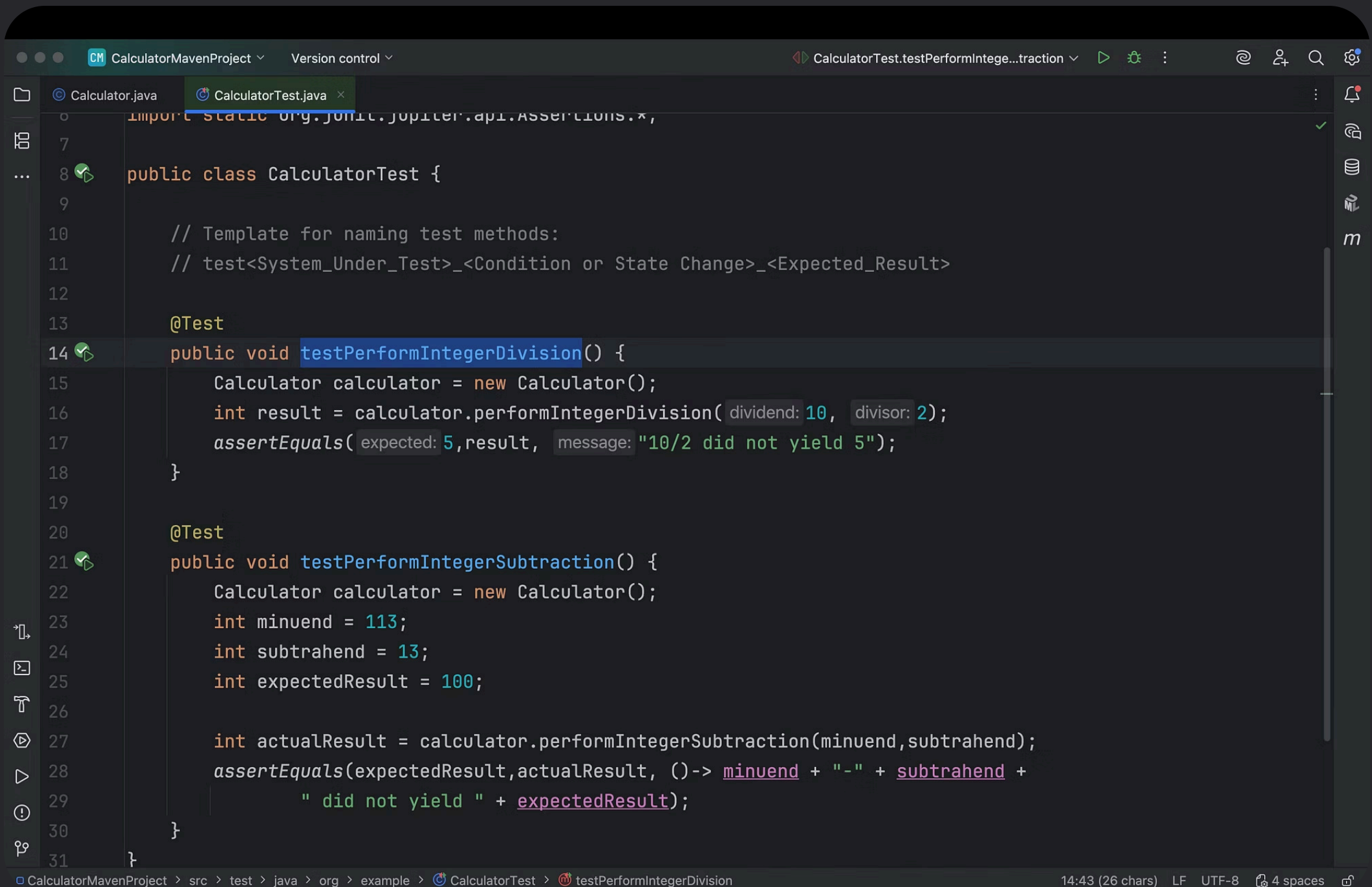
- **Example:**

```
testIntegerDivision_WhenDivisorIsValid_ThenReturnCorrectResult
```

- **Longer names** are **acceptable** because they **improve test comprehension**
- This **pattern** becomes especially **valuable** in large, **complex test suites**



# A template for naming unit test methods



```
import static org.junit.jupiter.api.Assertions.*;

public class CalculatorTest {

    // Template for naming test methods:
    // test<System_Under_Test>_<Condition or State Change>_<Expected_Result>

    @Test
    public void testPerformIntegerDivision() {
        Calculator calculator = new Calculator();
        int result = calculator.performIntegerDivision(dividend: 10, divisor: 2);
        assertEquals(expected: 5, result, message: "10/2 did not yield 5");
    }

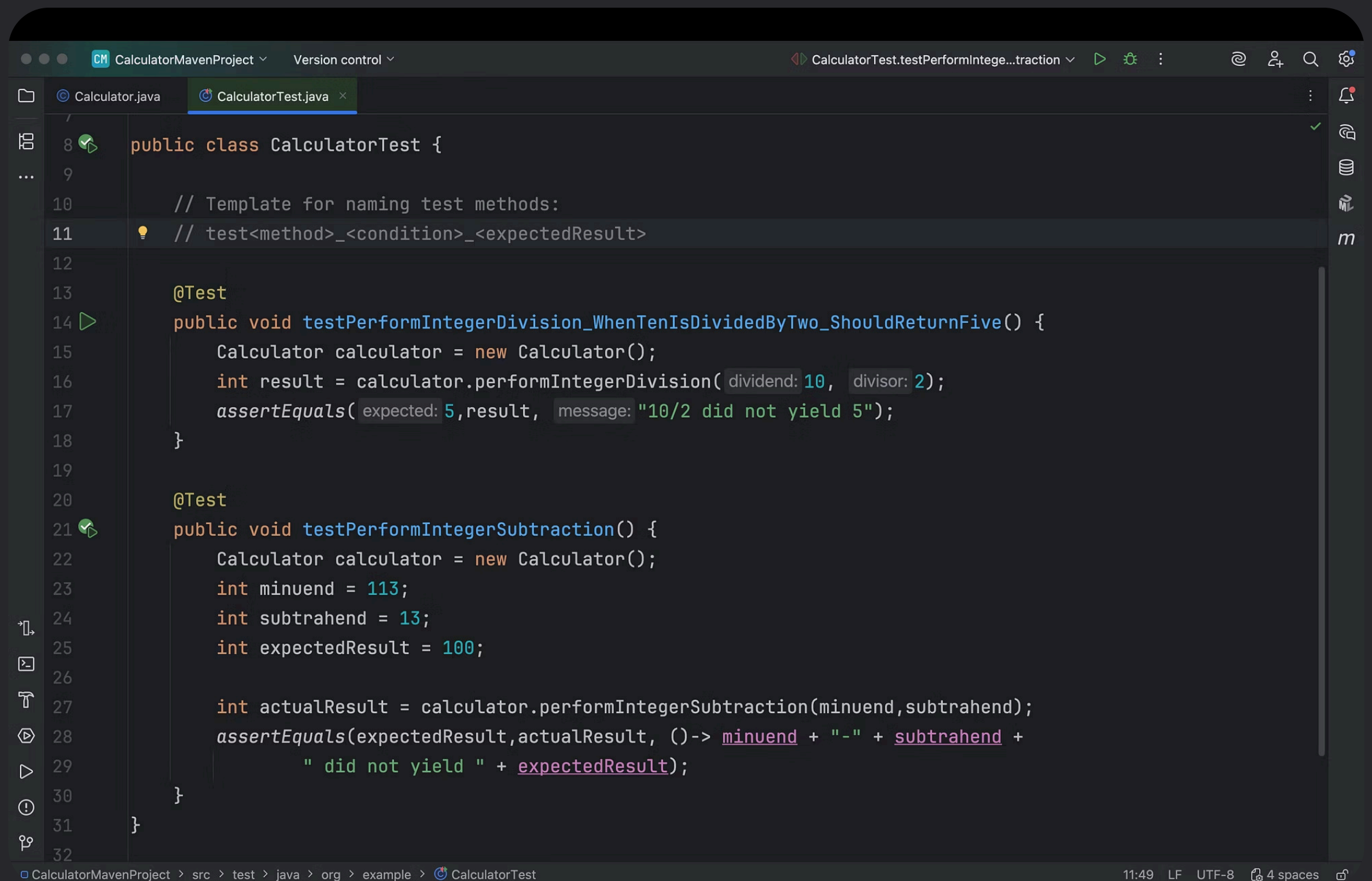
    @Test
    public void testPerformIntegerSubtraction() {
        Calculator calculator = new Calculator();
        int minuend = 113;
        int subtrahend = 13;
        int expectedResult = 100;

        int actualResult = calculator.performIntegerSubtraction(minuend, subtrahend);
        assertEquals(expectedResult, actualResult, () -> minuend + "-" + subtrahend +
            " did not yield " + expectedResult);
    }
}
```

**test<method>\_<condition>\_<expectedResult>**

# Refactor: renaming the unit test methods

Let's rename the first **test method** to be adherent with the **template for naming test methods**.

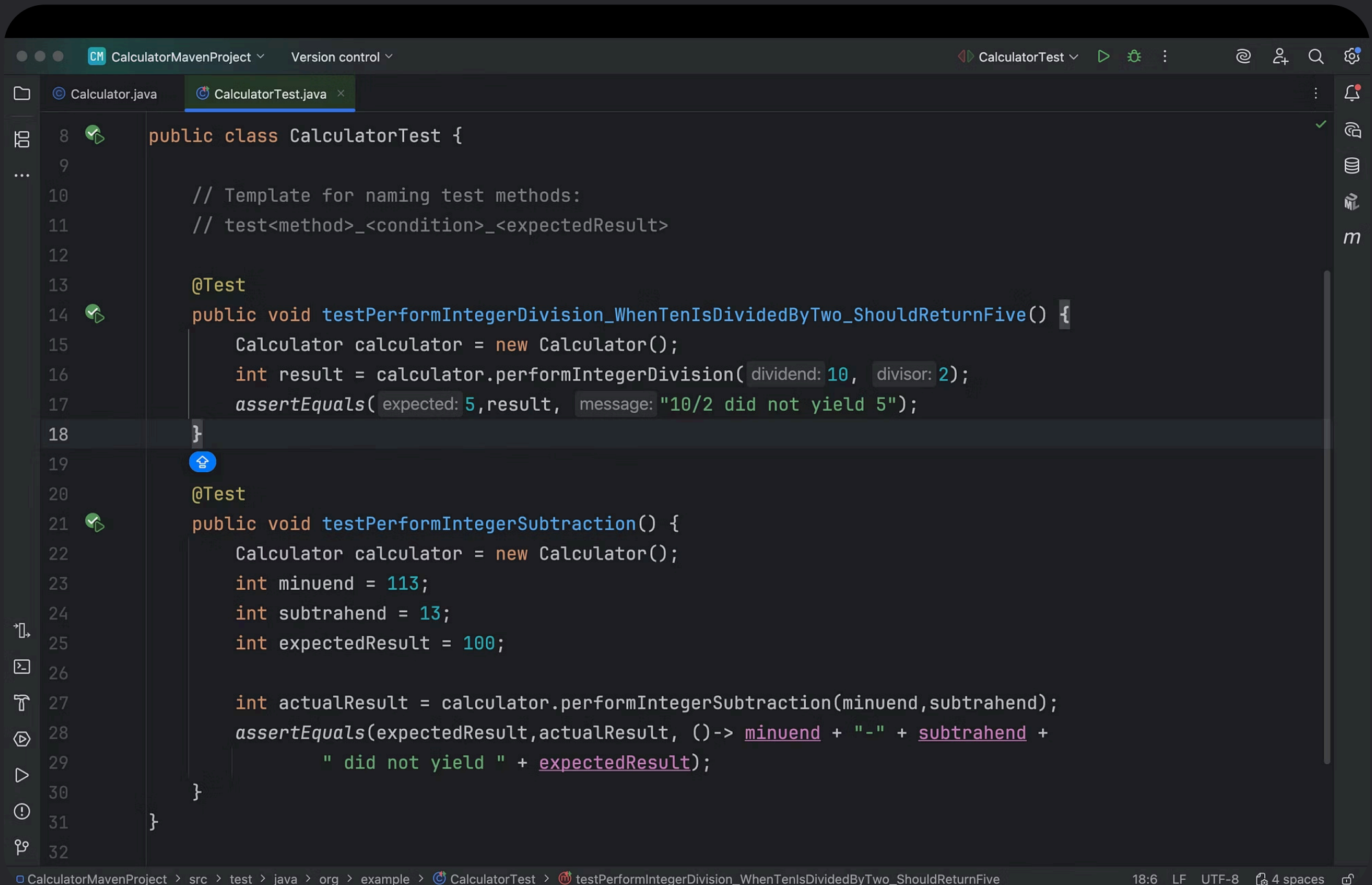


The screenshot shows an IDE window for a project named 'CalculatorMavenProject'. The 'CalculatorTest.java' file is open, showing a class with two test methods. The first method, 'testPerformIntegerDivision\_WhenTenIsDividedByTwo\_ShouldReturnFive()', is highlighted with a green checkmark icon. The second method, 'testPerformIntegerSubtraction()', is also highlighted with a green checkmark icon. The code is as follows:

```
8 public class CalculatorTest {
9
10     // Template for naming test methods:
11     // test<method>_<condition>_<expectedResult>
12
13     @Test
14     public void testPerformIntegerDivision_WhenTenIsDividedByTwo_ShouldReturnFive() {
15         Calculator calculator = new Calculator();
16         int result = calculator.performIntegerDivision(dividend: 10, divisor: 2);
17         assertEquals(expected: 5, result, message: "10/2 did not yield 5");
18     }
19
20     @Test
21     public void testPerformIntegerSubtraction() {
22         Calculator calculator = new Calculator();
23         int minuend = 113;
24         int subtrahend = 13;
25         int expectedResult = 100;
26
27         int actualResult = calculator.performIntegerSubtraction(minuend, subtrahend);
28         assertEquals(expectedResult, actualResult, () -> minuend + "-" + subtrahend +
29             " did not yield " + expectedResult);
30     }
31 }
32
```

The IDE interface includes a sidebar with icons for file explorer, search, and other tools. The bottom status bar shows the project path: 'CalculatorMavenProject > src > test > java > org > example > CalculatorTest', the time '11:49', and the encoding 'UTF-8'.

# Refactor: renaming the unit test methods

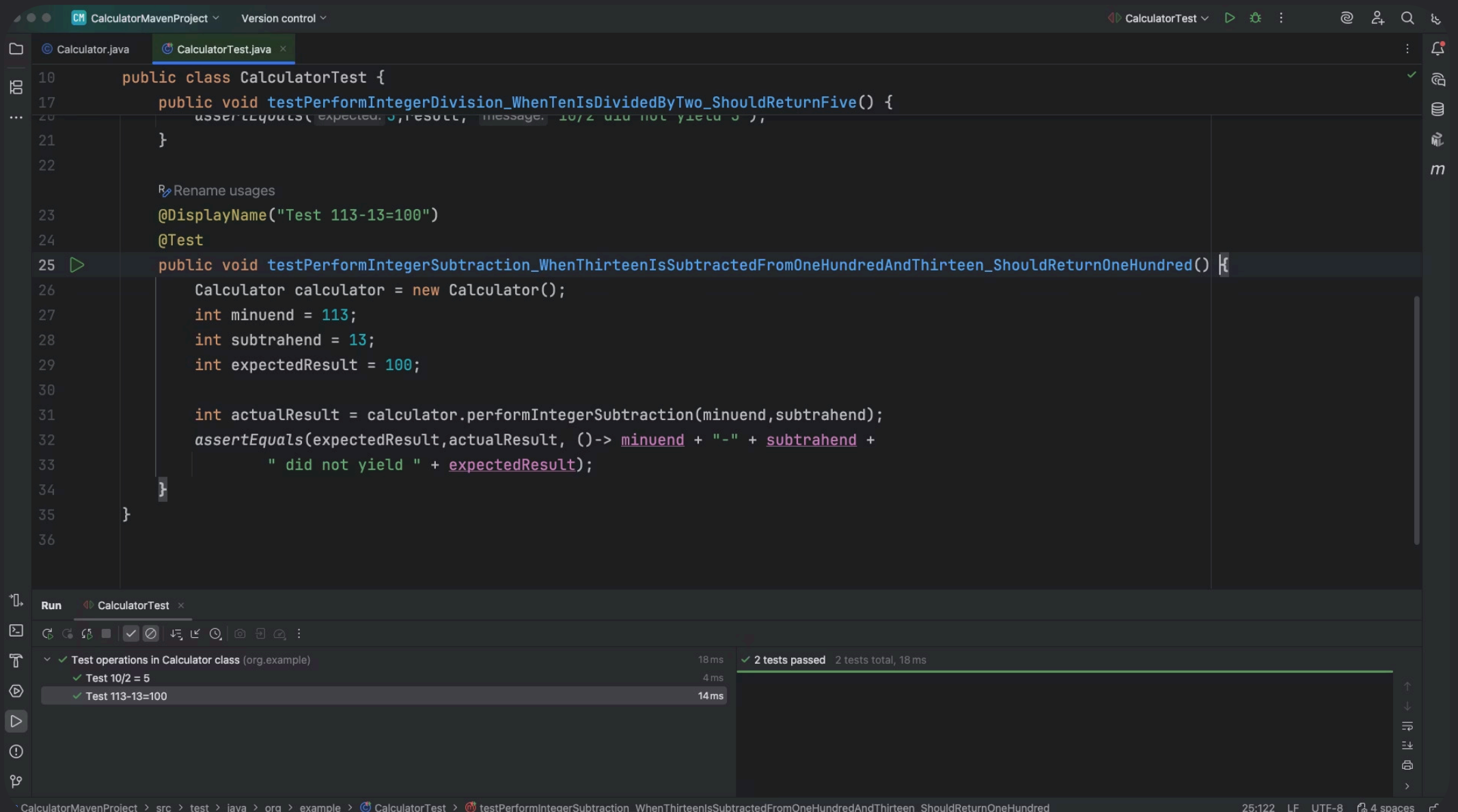


The screenshot shows an IDE window for a project named "CalculatorMavenProject". The "Version control" tab is active. The file "CalculatorTest.java" is open, showing the following code:

```
8 public class CalculatorTest {
9
10     // Template for naming test methods:
11     // test<method>_<condition>_<expectedResult>
12
13     @Test
14     public void testPerformIntegerDivision_WhenTenIsDividedByTwo_ShouldReturnFive() {
15         Calculator calculator = new Calculator();
16         int result = calculator.performIntegerDivision(dividend: 10, divisor: 2);
17         assertEquals(expected: 5, result, message: "10/2 did not yield 5");
18     }
19
20     @Test
21     public void testPerformIntegerSubtraction() {
22         Calculator calculator = new Calculator();
23         int minuend = 113;
24         int subtrahend = 13;
25         int expectedResult = 100;
26
27         int actualResult = calculator.performIntegerSubtraction(minuend, subtrahend);
28         assertEquals(expectedResult, actualResult, () -> minuend + "-" + subtrahend +
29             " did not yield " + expectedResult);
30     }
31 }
32
```

The IDE interface includes a sidebar with icons for Explorer, Search, Run and Debug, and other tools. The bottom status bar shows the project path: "CalculatorMavenProject > src > test > java > org > example > CalculatorTest > testPerformIntegerDivision\_WhenTenIsDividedByTwo\_ShouldReturnFive". The status bar also displays "18:6 LF UTF-8 4 spaces".

# Refactor: renaming the method to test the Calculator's performIntegerSubtraction method



The screenshot shows an IDE with a Java project named 'CalculatorMavenProject'. The 'CalculatorTest.java' file is open, showing a test class with two methods. The second method, 'testPerformIntegerSubtraction\_WhenThirteenIsSubtractedFromOneHundredAndThirteen\_ShouldReturnOneHundred()', is highlighted. Below the code editor, the 'Run' tab shows the execution results for 'CalculatorTest'. The results indicate that 2 tests passed: 'Test 10/2 = 5' (4 ms) and 'Test 113-13=100' (14 ms). The total execution time was 18 ms.

```
10 public class CalculatorTest {
17     public void testPerformIntegerDivision_WhenTenIsDividedByTwo_ShouldReturnFive() {
20         assertEquals(expected: 5, result, message: "10/2 did not yield 5",
21     }
22
23     @DisplayName("Test 113-13=100")
24     @Test
25     public void testPerformIntegerSubtraction_WhenThirteenIsSubtractedFromOneHundredAndThirteen_ShouldReturnOneHundred() {
26         Calculator calculator = new Calculator();
27         int minuend = 113;
28         int subtrahend = 13;
29         int expectedResult = 100;
30
31         int actualResult = calculator.performIntegerSubtraction(minuend, subtrahend);
32         assertEquals(expectedResult, actualResult, () -> minuend + "-" + subtrahend +
33             " did not yield " + expectedResult);
34     }
35 }
36
```

Run CalculatorTest

Test operations in Calculator class (org.example)

| Test Name       | Duration |
|-----------------|----------|
| Test 10/2 = 5   | 4 ms     |
| Test 113-13=100 | 14 ms    |

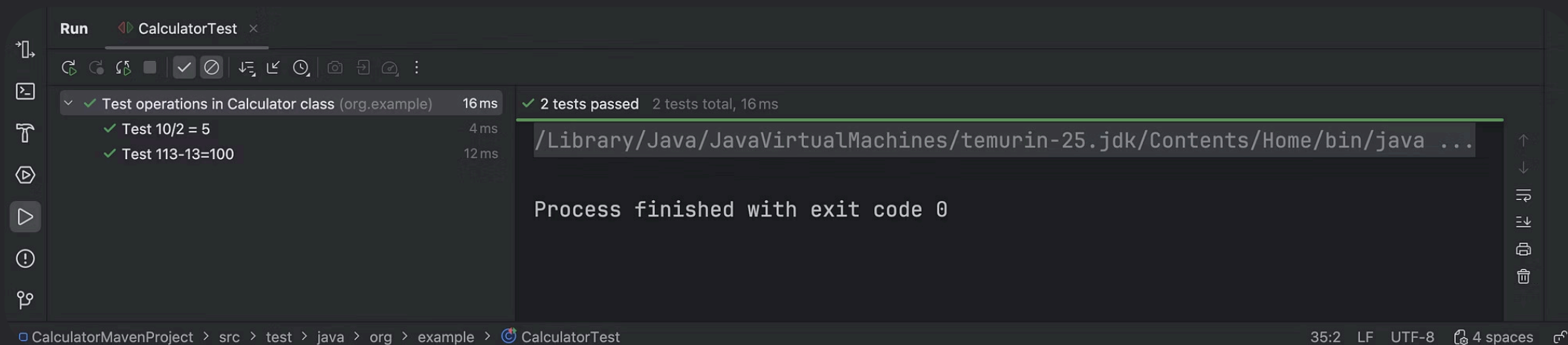
2 tests passed 2 tests total, 18 ms

CalculatorMavenProject > src > test > java > org > example > CalculatorTest > testPerformIntegerSubtraction\_WhenThirteenIsSubtractedFromOneHundredAndThirteen\_ShouldReturnOneHundred

25:122 LF UTF-8 4 spaces

testPerformIntegerSubtraction\_WhenThirteenIsSubtractedFromOneHundredAndThirteen\_ShouldReturnOneHundred

# Run all tests again



They should **pass!**

# Benefits and Drawbacks of the Naming Test Methods Template

## Benefits

- **Benefits:** readability, maintainability, and easier collaboration
- Helps new team members quickly understand intent without reading code
- Reduces ambiguity by encoding conditions and expected behavior directly in the name

## Drawbacks

- **Main drawback:** names may become long in test reports
- Later improvements (e.g., display names) can mitigate this issue



# Conclusion

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Clear and descriptive test method names are crucial for maintainable test suites

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The naming pattern helps encode scenario + behavior directly into the method name

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Even if verbose, this approach improves collaboration and long-term clarity

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Future refinements (like display names) help balance readability in reports





# Open Floor Discussion

- Do you currently follow a naming convention for your test methods?
- How long is "too long" for a test name in your team's context?
- Would descriptive names speed up onboarding or code review in your projects?